

Examiner reports

Q1.

Part (a) was generally answered well. There were however some curious assumptions, for example “the ball has no mass”, which were not appropriate. Part (b) was generally done quite well, with candidates obtaining the results without too much difficulty. Part (c) proved to be more demanding, but candidates often tended to do very badly or very well. If they could produce a correct equation for t they could often gain all the marks, but other candidates could not see how to form an equation that would enable them to start the question. In a few cases the candidates rounded the time of flight and then obtained an incorrect value for the range.

Q2.

The candidates seemed to find the fact that the arrow was fired horizontally difficult to handle. Many appeared to try to give the arrow a vertical component of velocity. In part (a) many candidates seemed to have been helped by the printed answer. In some cases they included an unnecessary negative sign, which led to a negative time. There were some good answers to part (b), but some candidates included an acceleration g . There were very varied answers to part (c). Some candidates gave very good responses, but there were very many confused ones too, which revealed a lack of understanding of the principles of this topic.

Q3.

Part (a) was often done well, but a number of candidates effectively repeated the same assumption twice. The candidates were often helped by the printed answer in part (b). Many of the candidates made life difficult for themselves by finding the initial speed and angle of projection and then working with these values. One consequence of this was that the final answers were inaccurate. Part (c) proved more challenging for candidates. The most common incorrect approach was to assume that the ball hit the wall when it was at its maximum height. In part (d) there were some good answers and many of the candidates were on the right track. However quite a few of the candidates produced answers which lacked clarity of explanation.

Q4.

The vast majority of the candidates answered part (a) of the question correctly.

For part (b), almost all candidates substituted 100 for x and 4 for y . However, some candidates had difficulty simplifying the resulting equation.

For part (c), most candidates stated that they assumed the ball was a particle, there was no air resistance or that the only force acting on the ball was the Earth's gravitational force. However, there were some candidates who stated the assumption of the ball being a particle without mass!

Q5.

This question proved more discriminating than expected.

Q6.

Candidates produced good work in this question, and were able to score marks on the parts where they had the necessary knowledge and understanding. Part (a)(i) was done well, and part (a)(ii), although an unfamiliar request, produced many good solutions. Part (a)(iii) was done well in a variety of ways. Parts (b) and (c) proved successful for most candidates.

Q7.

It was refreshing to see a wide variety of approaches to this question, which differed in many ways from previous questions on this topic. Many candidates were able to find the coordinates of the ball in part (a) and at this stage an error in the direction of the y-value was condoned. Part (b) provided interesting responses, using either the answers to part (a) with subsequent elimination of the time, or, in some cases, the full equation of the trajectory. In part (c) many candidates used the result of part (b), while others successfully found the time involved as an intermediate step. Part (d) proved to be more discriminating and there was confusion, for some, as to the quantities involved in the motion in the two directions. Many candidates reached the correct result following sound work, but some lost a mark through failing to adhere to the degree of accuracy required by the rubric for the paper. Overall it is encouraging to see candidates who are unable to complete one part of a question continue to other parts where they can be more successful.

Q8.

Parts (a) and (b) were done well by the better prepared candidates, but few spotted and used the pattern of decreasing distances in part (c).

Q9.

The quality of solutions to problems set on this area of the specification has improved steadily since the first paper in 2001. It was pleasing to see a variety of methods being used and many candidates scored well on this demanding question.

Part (a) was attempted successfully either using the equations of constant acceleration or by integration. The most common error was an incorrect sign in the acceleration of the vertical component.

Part (b) proved to be the most demanding part of the question. Again, varied methods were used, ranging from a full verification of the result to the quotation of the general equation of the path of the trajectory with appropriate substitutions.

Part (c) was popular, yielding marks to most candidates.

The varied methods used in part (d) revealed clear understanding of the motion. The most popular methods used the result quoted in part (b) or the equations of motion with constant acceleration, sometimes incorporating the symmetry of the motion.

In part (e), for those candidates who realised that the greatest speed occurred either at O or at C , the request was straightforward and full marks were often obtained.

Q10.

This last question revealed an improved understanding of projectile motion, as well as

clear and accurate arithmetic work. A variety of methods were used in all parts of the question, including use of the symmetry of the motion.

Part (a) was done well, with vertical motion being considered both upwards and downwards. In part (b), again both directions of the vertical motion were used. Part (i) was done well, but part (ii) proved more discriminating, with a weakness in solving quadratic equations being shown by some. Part (iii) was again mostly successfully and accurately completed.

Q11.

Again the poor use of algebra proved to be the downfall of many candidates. Part (a) was answered quite well, with a satisfactory proportion of candidates reaching the printed solution, but there were many careless errors in part (b). Part (c) was generally successful, but few were able to apply the same principle in part (d), with the given speed often being treated as a vertical velocity component.

Q12.

The printed answer helped candidates to do well in part (a). A few candidates did not show enough working to justify the allocation of all three marks. There were some candidates who worked with equations for the vertical component of the motion, rather than the horizontal component.

Part (b) was again well answered. A few candidates made arithmetic errors and some found the maximum height instead of the height of the crossbar.

Candidates found part (c) very difficult. Many of them considered the position of the ball rather than its velocity. Some considered only one component of the velocity. However there were a few candidates who did produce a good solution to this part of the question. Most of the candidates stated appropriate assumptions, although some candidates produced inappropriate assumptions.

Q13.

Part (a) was done well by many candidates, who may have been helped by the printed answer. There were also some candidates who made little progress and did not seem to be able to see how to form an equation to find the time. Part (b) was often done well, but was completed by fewer candidates than part (a). Some candidates did not seem to be able to form a correct expression for the height. A few candidates used

$$h = 10 \sin 60^\circ t + \frac{1}{2} g t^2$$
 instead of
$$h = 10 \sin 60^\circ t - \frac{1}{2} g t^2$$
. Part (c) proved to be very challenging.

Many candidates made very little progress with this part of the question. Some found the vertical component of the velocity when the ball hit the wall. Only a small proportion of the candidates could incorporate the horizontal component and find the correct speed.

Q14.

The candidates who knew how to tackle projectile questions with confidence did well with this question. Part (a) did not present much difficulty. However, in part (b), including the correct signs and solving the quadratic equation correctly were more challenging tasks.

There were some very poor attempts at solving the quadratic equation, where candidates did not realise that the quadratic equation formula was required.

Q15.

There were a fair number of good responses to part (a). The candidates who did not produce a correct solution generally either missed out the difference between the initial and final heights of the ball, or made sign errors in the setting up of their equations. A few candidates made arithmetic errors. Part (b) was also done well, with some candidates being helped by the printed answer in part (a). A few candidates used incorrect times from part (a) when trying to find the range. There were only a few correct solutions to part (c). An incomplete solution, which only found the vertical component of the velocity, was quite common among those who made some progress with the solution.

Q16.

Part (a) was answered quite well, with many candidates making good use of the printed answer.

Part (b) caused more problems. Some candidates only found the vertical component of the velocity and did not bring the horizontal component into their working. As a result there were relatively few correct answers for the speed, even though quite a few candidates found the vertical component correctly. There were also quite a number of candidates you did not really know how to use the height to find a time or a velocity.

Q17.

Part (a) was either done very well or very badly. Some candidates produced simple, correct arguments. Two common incorrect approaches were to take $v = V \sin \alpha$ instead of $u = V \sin \alpha$, or to omit the negative sign from the acceleration when using the equation as $v^2 = u^2 + 2as$. The other approach that caused difficulty was when candidates tried to find

the time for the maximum height by working with the expression $y = V \sin \alpha t - \frac{1}{2}gt^2$. This led to a variety of problems. The candidates who used the formula from part (a) usually had no difficulty finding V . A number of candidates did not use this result and often ended up with the vertical component of the initial velocity instead of the required speed. Part (c) caused problems for some candidates, the biggest problem being finding the time of flight correctly. Some used an equation that would have given the time for the maximum height by considering $y = 6$.

Q18.

While there were some good proofs of the formula for the range, there were also a number of confused approaches. Several of these were based on the formula as $v^2 = u^2 + 2as$. However, those who found the time of flight first often made quite good progress. Some candidates misquoted the double angle formula. Another area of difficulty was in the solution of the quadratic equation to find the time of flight.

In part (b), almost all of the candidates gave the answer 45° , but very few were able to give a convincing argument.

Part (c) was generally done very well. Some candidates tried to do this without using the

results from parts (a) and (b). Many of these candidates were unsuccessful. A few candidates did not give their answer for the range to the nearest metre as instructed in the question.

Q19.

There were a number of good solutions to part (a). Some candidates did make sign errors when setting up the equation for the vertical motion. Some candidates broke the motion down into stages, but those that did often made small errors. Part (b) was done very well, with some candidates using the time given in part (a) to enable them to find the range. Part (c) proved to be very demanding. Some candidates calculated the vertical component of the velocity when the car hit the sea, but did not bring the horizontal component into their solutions. There were only very few complete solutions.



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